

Keshav Ajagaonkar

📞 +91 9359545731 ✉ keshavvamanajagaonkar@gmail.com 🔗 LinkedIn 🐙 GitHub 🚀 LeetCode 🧑🏻 GeeksforGeeks

PROFESSIONAL SUMMARY

Final-year Computer Engineering student with strong fundamentals in data structures and algorithms. Built production-grade full-stack applications featuring AI-driven automation and real-time communication. Passionate about engineering scalable software solutions for enterprise-scale challenges.

EDUCATION

Terna Engineering College, Mumbai University
Bachelor of Engineering in Computer Engineering

Expected May 2026
CGPA: 8.72 / 10 (Sem I – VII)

TECHNICAL SKILLS

Languages: C++, C, JavaScript, Python, Java, SQL

Core CS: Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks

Frontend: React.js, Redux Toolkit, Tailwind CSS

Backend: Node.js, Express.js, MongoDB, Mongoose, SQL Databases, REST API Design

Real-Time & Cloud: Socket.io, WebRTC, AWS S3, Cloudinary, Nodemailer

AI Integration: Google Gemini API, NLP Classification, Sentiment Analysis, Automated Categorization

Auth & Security: JWT, Role-Based Access Control, Zod Schema Validation, Input Sanitization

Tools : Git, GitHub, VS Code

PROJECTS

CivicMitra – AI-Powered Civic Grievance Platform

Sep 2025 – Present

Tech: React 19, Node.js, Express 5, MongoDB, Socket.io, Gemini AI, JWT, Zod

[\[GitHub\]](#)

- Architected a multi-role enterprise system (Citizen, Staff, Worker, Admin) with 30+ REST endpoints, 8 database models, JWT authentication, and role-based middleware – managing the full complaint lifecycle from submission to resolution.
- Engineered a two-stage Gemini AI pipeline that auto-classifies complaints into 9 categories with department routing, generates NLP summaries (sentiment, urgency, key points), and flags semantic duplicates – with a rule-based fallback ensuring 100% service availability.
- Designed a crowdsourced priority scoring algorithm using a weighted formula combining upvotes, urgency, time decay, and AI-derived sentiment, alongside a geospatial community feed powered by MongoDB 2dsphere indexes.
- Achieved 85%+ automated test coverage using Jest and Supertest, optimized database performance with 7 compound indexes, and implemented real-time Socket.io notifications broadcasting status updates to all complaint supporters.

Relic – Content-Addressable Version Control System

Jan 2026 – Feb 2026

Tech: Node.js, SHA-256, zlib, AWS S3, Yargs, fs Module

[\[GitHub\]](#)

- Built a version control system from scratch implementing Git's core architecture: a content-addressable object store where every blob, tree, and commit is SHA-256 hashed and zlib-compressed, enabling automatic deduplication of identical content.
- Implemented the complete object model – blobs for file content, tree objects mapping filenames to blob/subtree hashes for directory snapshots, and commit objects forming a DAG-based history with parent links, author metadata, and timestamps.
- Developed 8 modular CLI commands (init, add, commit, log, diff, branch, checkout, revert) parsed through Yargs, demonstrating understanding of low-level file system operations and data structure design.
- Integrated AWS S3 for remote repository synchronization via `putObject/getObject/listObjectsV2`, enabling cross-machine push and pull of the compressed object store and ref state.

PulseChat – Real-Time Communication Platform

Jul 2025 – Dec 2025

Tech: React 19, Node.js, Express 5, MongoDB, Socket.io, WebRTC, Cloudinary

[\[GitHub\]](#)

- Developed a scalable real-time messaging platform supporting one-on-one and group conversations with admin controls, threaded replies, emoji reactions, and file sharing – managing 10+ Socket.io event types via room-based channels.
- Implemented peer-to-peer audio and video calling using WebRTC with full ICE candidate exchange, SDP offer/answer negotiation, and server-side call state tracking for busy and offline handling.
- Designed a three-stage message receipt system (sent, delivered, read) using Socket.io for instant front-end updates combined with MongoDB batch writes to persist delivery status reliably across all chat types.
- Built a media pipeline through Cloudinary with MIME-based routing for images, videos, and documents, server-side transformations for profile avatars, and client-side blob downloads with error fallback.

ACHIEVEMENTS

- Ranked in the **top 0.1%** statewide in HSC 2021 with a score of 95.5%.
- Secured **89% in SSC 2019** (Maharashtra State Board) with distinction.
- Earned distinction in every semester of B.E. Computer Engineering (Sem I through VII).

CERTIFICATIONS

- Full Stack Web Development – [Apana College](#)

2025